

CNG 101

A Beginner's Guide to Compressed Natural Gas

Everything you need to know about switching to CNG —
save up to 50% on fuel costs across Nigeria

50%

Fuel cost savings

!1,200

Avg. price per kg CNG

3 Cities

Lagos · Ilorin · Abuja

CHAPTER 1 — WHAT IS COMPRESSED NATURAL GAS (CNG)?

Compressed Natural Gas (CNG) is natural gas — primarily methane (CH₄) — that has been compressed to less than 1% of its volume at standard atmospheric pressure. CNG is stored and distributed in hard, sealed cylindrical or spherical containers at pressures typically between 200–248 bar (3,000–3,600 psi).

In Nigeria, CNG is rapidly emerging as the most practical alternative to petrol (PMS) and diesel. With the Federal Government's commitment to expanding CNG infrastructure and companies like Adesa Energy rolling out mobile conversion and refueling services, switching to CNG has never been easier.

CNG is not the same as LPG (cooking gas). CNG is methane; LPG is propane/butane. They have different tanks, pressures, and vehicle fittings.

CHAPTER 2 — WHY SWITCH TO CNG?

The case for CNG in Nigeria is compelling across three dimensions: cost, environment, and energy security.

Factor	
Average price (2026)	₦897–₦1,200 / litre PMS vs ₦1,200 / kg CNG
Energy equivalent	1 litre PMS ≈ 0.72 kg CNG CNG often cheaper
Fuel cost savings	Up to 50% lower running costs per kilometre
Engine wear	CNG burns cleaner — less carbon deposit, longer engine life
CO ₂ emissions	~20–25% less CO ₂ than petrol; near-zero particulates
Octane rating	CNG has octane rating of ~130 vs ~91 for regular PMS
Range anxiety	Bi-fuel vehicles keep petrol tank as backup
Infrastructure (Lagos)	Growing — Adesa Energy mobile refueling eliminates

range anxiety

A Lagos commercial bus driver covering 150 km/day typically saves ₦15,000– ₦22,000 per month after switching to CNG with Adesa Energy.

CHAPTER 3 — HOW DOES CNG CONVERSION WORK?

Converting your vehicle to CNG means fitting a bi-fuel kit — a system that allows your engine to run on either CNG or petrol at the flick of a switch. The vehicle retains its original petrol tank and functionality while gaining a CNG cylinder and injection system.

The conversion kit has five core components:

- CNG Cylinder — A high-pressure steel or composite tank fitted in the boot or under the chassis. Capacities range from 40L to 120L water-volume equivalent.
- High-Pressure Regulator — Reduces cylinder pressure (~200 bar) to engine-usable pressure (~1–8 bar) in stages.
- Sequential Gas Injection (SGI) System — Injects CNG into the intake manifold in sync with each cylinder's firing cycle, mimicking petrol injection.
- Electronic Control Unit (ECU) — Monitors engine parameters, switches between fuels, and optimises the air-fuel mixture for CNG.
- Fuel Selector Switch — Mounted on the dashboard; lets the driver choose CNG, petrol, or automatic (prioritises CNG when cylinder pressure is adequate).

CHAPTER 4 — THE ADESA ENERGY CONVERSION PROCESS

Adesa Energy performs mobile conversions at your location — no need to drive to a workshop. Here is what to expect:

1. Booking — Join our waitlist at www.adesaenergy.com/waitlist or call +234 816 882 3730 to schedule your conversion appointment.
2. Vehicle Assessment (30 min) — Our technician inspects your engine, fuel system, and chassis to confirm suitability and recommend the right kit size.
3. Kit Installation (3–5 hours) — We mount the CNG cylinder, run high-pressure lines, install the SGI ECU, connect injectors, and calibrate the system.
4. Leak & Safety Test — Every connection is tested with a calibrated gas-leak detector before the vehicle leaves our care.
5. Certification & Documentation — You receive a conversion certificate, kit serial numbers, and a maintenance schedule.
6. First Fill & Demo — Your first CNG fill is done on-site. Our technician shows you how to operate the fuel-selector switch and read the CNG gauge.

The entire process takes a single day. Adesa Energy offers a 12-month warranty on all installed kits and free post-installation check-up at 500 km.

CHAPTER 5 — WHICH VEHICLES CAN BE CONVERTED?

Most petrol-powered vehicles with electronic fuel injection (EFI) — made from 2000 onwards — are compatible. The table below gives a quick guide:

Vehicle Type		
Petrol EFI cars (2000+)	' Fully compatible	Toyota, Honda, Hyundai, Kia, Nissan
Carbureted petrol cars	& p Possible — older kit	Pre-2000 Peugeot, VW Beetle
Commercial buses (EFI)	' Fully compatible	Hiace, Sprinter, Mini-bus
Trucks / HGVs (petrol)	' Large-cylinder kits	Customised per vehicle
Diesel engines	'L Not supported	Diesel requires separate CNG-diesel kit
CNG-only (OEM) vehicles	N/A — already CNG	Some Hyundai, Toyota fleet models

CHAPTER 6 — CNG SAFETY: FACTS VS MYTHS

MYTH: CNG tanks explode easily

FACT: CNG cylinders are tested to 3× working pressure and made from steel or composite materials that are far tougher than petrol tanks. They're designed to vent gas upward — not pool liquid — in the event of a leak.

MYTH: CNG is dangerous in an accident

FACT: Because CNG is lighter than air, it disperses rapidly rather than pooling on the ground like petrol. In crash tests, CNG tanks consistently outperform petrol tanks.

MYTH: You'll run out of gas with no warning

FACT: The dashboard gauge shows CNG level, and the ECU automatically switches to petrol when the cylinder pressure drops, giving you plenty of range to reach the next fill point.

MYTH: CNG reduces engine performance

FACT: CNG's higher octane (130) can actually improve combustion efficiency. Some vehicles gain a slight power advantage on CNG, particularly in city driving conditions.

CHAPTER 7 — CNG MAINTENANCE CHECKLIST

CNG systems require minimal maintenance, but the following schedule keeps your kit in peak condition:

Every 500 km (first 1,000 km)

Post-installation leak inspection by Adesa technician (free)

Every 10,000 km

Regulator and filter inspection; replace inline filter if needed

Every 20,000 km

Full system pressure test; ECU re-calibration if required

Every 3 years

Mandatory cylinder hydrostatic re-test (required by DPR/NMDPRA)

Every 10–15 years

Cylinder replacement (manufacturer-specified service life)

After any accident

Immediate professional inspection before using CNG fuel again

Ready to Switch to CNG?

Join our waitlist today and lock in early-bird conversion pricing.
Mobile service — we come to you anywhere in Lagos, Ilorin, or Abuja.

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